

Trustworthy Artificial Intelligence in Adversarial Environments

Adversarially Robust, Privacy-Aware, and Computationally Efficient Learning Systems

1. Objective

Problem I

Adversarial Robustness of Learning Systems

Formal Guarantees under Deliberate Perturbations

Problem II:

Robustness-Accuracy-Privacy Trade-off

Joint Optimisation of Competing Objectives

2. Master Problems

(S1) Continuous-Time Temporal Graph Modelling

(S2) Multi-Granularity Representation Learning

(S3) Post-Quantum Secure Architectures

(S4) Zero-Shot and Out-of-Distribution Generalisation

(S5) Stochastic Adversarial Optimisation

(S6) Privacy-Preserving Distribution Alignment

(S7) Efficient Sequential Inference Mechanisms

(S8) Online Concept Drift Adaptation

4. Sub-Problems

(M1) CT-TGNN

98.3% Accuracy

(M2) TripleE-TGNN

96.8% Accuracy

(M3) FedLLM-API

87.1% Byzantine Robustness

(M4) PQ-IDPS

96.2% Detection Rate

(M1)

98.3% Accuracy

(M2) TripleE-TGNN

96.8% Accuracy

(M3) FedLLM-API

87.1% Byzantine Robustness

(M4) Game-Theoretic Ralaa

94.7% Strategic Stability

4. Key Methodologies

(M1) CT-TGNN

98.3% Accuracy

(M2) TripleE-TGNN

96.9% Accuracy

(M4) PQ-IDPS

96.2% Detection Rate

(M7) Game-Theore

94.7% Strategic Stability

Unified Optimisation Framework (Algorithm 8):

Alternating Optimisation with Consistency Regularisation

Theoretical Convergence Rate: $O(1/\sqrt{T})$ for Coupled Multi-Method System

Mathematical Foundations: Neural Ordinary Differential Equations · Graph Theory · Optimal Transport · State-Space Monte Carlo